

A COMPILED PRACTICE SET

Chemistry

Mixed-topic practice paper

19

MARKS

35

MINUTES EST.

3SOURCE PAPERS

Instructions

Show your working and write answers in the spaces provided.
Questions have been selected and renumbered for this mixed set.
Solutions are not included in this demo paper.

Questions

QUESTION 1	Solubility and equilibrium Cheltenham Girls High School - 2022 Chemistry Trial (Question 21)	8 marks
QUESTION 2	Acid-base titration and thermochemistry Girraween High School - 2022 Chemistry Trial (Question 25)	6 marks
QUESTION 3	Acid-base equilibrium James Ruse Agricultural High School - 2022 Chemistry Trial (Question 22)	5 marks

Each question page includes a direct link to its original source paper.
This demonstration shows how selected question pages can be assembled into a reusable set.

Question 1

8 MARKS

Solubility and equilibrium

Source: Cheltenham Girls High School | 2022 Chemistry Trial | Question 21

SOURCE PDF

- (a) Compare and contrast the relative solubility of lead (II) hydroxide and lead (II) phosphate.

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Question 1

8 MARKS

Solubility and equilibrium

Source: Cheltenham Girls High School | 2022 Chemistry Trial | Question 21

SOURCE PDF

- (b) The solubility product, K_{sp} of $\text{Ca}(\text{OH})_2$ is 6.4×10^{-6} at 298K.

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Calculate the solubility (in g L^{-1}) of calcium hydroxide in pure water at 298K.

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- (c) Using the information below, explain the effect a temperature increase would have on the quoted K_{sp} .

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Thermodynamic data (at 25 °C, 100 kPa)

	$\Delta H^\circ_f / \text{kJ mol}^{-1}$
$\text{Ca}^{2+} (\text{aq})$	-543
$\text{OH}^- (\text{aq})$	-230
$\text{Ca}(\text{OH})_2 (\text{s})$	-986

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Question 2

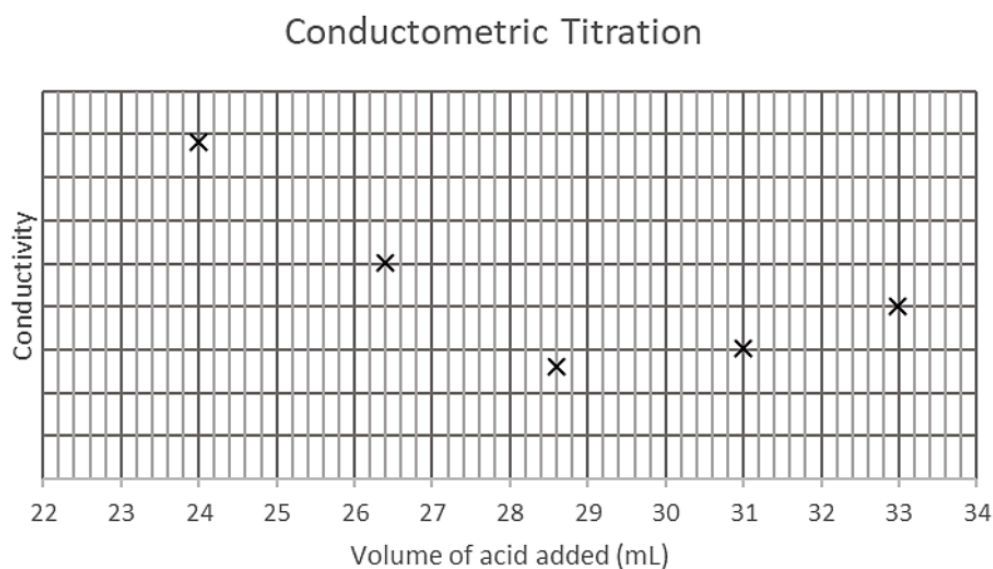
6 MARKS

Acid-base titration and thermochemistry

Source: Girraween High School | 2022 Chemistry Trial | Question 25

SOURCE PDF

A conductometric titration was carried out using 25.00 mL of barium hydroxide and 0.100 mol^{-1} sulfuric acid. Conductivity measurements were recorded at five different intervals and the results plotted on the graph below.



- a) Using information from the graph, calculate the concentration of the barium hydroxide solution. Include a suitable chemical equation in your answer.

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Question 2

6 MARKS

Acid-base titration and thermochemistry

Source: Girraween High School | 2022 Chemistry Trial | Question 25

SOURCE PDF

- b) The titration was performed in a thermally insulated vessel. The sulfuric acid and barium hydroxide solutions were 20.2°C at the start of the titration. 3
The enthalpy of neutralisation is $-57.6 \text{ kJ mol}^{-1}$.

Calculate the final temperature of the solution when 40.0 mL of the acid is added to 25.00 mL of the base.

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Question 3

5 MARKS

Acid-base equilibrium

Source: James Ruse Agricultural High School | 2022 Chemistry Trial | Question 22

SOURCE PDF

Ammonia is a weak base with a K_b value of 1.8×10^{-5} .

Determine the pH of a 0.500 molL^{-1} ammonia solution?

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